University of Florida, Department of Mathematics

Topology First-Year Exam, January 2022, Part 2

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that the torus T is not homeomorphic to the surface of genus 2, $M_2 = T \# T$.

2. Prove that the set of irrational numbers $\mathcal{J} \subset \mathbb{R}$ cannot be presented as a countable union of closed sets each of which has empty interior in $\mathcal{J}$.

3. Prove that $\pi_1(S^1) = \mathbb{Z}$.

4. Let CX denote the cone over the n-point space $X = \{1, \dots, n\}$. Show that every continuous map $f : CX \to CX$ has a fixed point.

5. Show that the fundamental group of $M = K \# T$ is not abelian where K is the Klein bottle.

Answer the following with complete definitions or statements or proofs.

6. Show that every map $f : S^2 \to \prod_{\alpha \in J} S^1$ of the 2-sphere to the product of circles is null-homotopic.

7. Show that for a retraction $r : X \to A$ the induced map $r_\#$ on fundamental groups is surjective.

8. Which of the following spaces taken with the product topology are locally compact? separable? Here $I = [0, 1]$ and $\mathbb{N}$ is the set of natural numbers.

(a) $\mathbb{N}^{\mathbb{N}}$;
(b) $I^{\mathbb{N}}$;
(c) $\mathbb{N}^I$;
(d) I^I.

9. State the Seifert–van Kampen Theorem. Can it be used to compute the fundamental group of S^1?

10. Show that S^2 is not homeomorphic to S^4.